NeuroFlow — Methods (compile check)

Methodology

Study population and 4D Flow MRI acquisition

Seven patients with cerebrovascular pathology were prospectively enrolled and scanned on two MRI platforms within a single imaging session: a clinical 3 T Siemens system and an investigational 7 T Siemens system. For each subject, paired 4D Flow MRI studies were acquired on both scanners to minimise inter-session physiological variability. Each acquisition produced four co-registered volumetric time series — a magnitude image and three directional phase-contrast velocity maps encoding velocity along the feet–head (u), left–right (v), and anterior–posterior (w) axes [?, ?]. The 3 T series covered between nine and fourteen cardiac frames, while the 7 T series spanned ten to fifteen frames, all encoded with a common velocity-encoding sensitivity of VENC = 0.9 m/s across both field strengths [?]. Raw data were converted from DICOM to NIfTI, preserving the original phase representation and applying the LPS → RAS sign convention so that the three velocity components retained their physical orientation in the scanner frame.

Figure 1: Acquisition and pairing overview. Paired 3 T and 7 T 4D Flow MRI acquired within a single imaging session for each subject.

Motion correction and multi-field-strength registration

Cardiac-gated 4D Flow MRI is intrinsically sensitive to bulk motion between cardiac phases. Intra-scan motion was corrected by estimating rigid-body transformations $\{\mathbf{T}_t^{\text{rigid}}\}_{t=1}^{T}$ from the magnitude time series relative to a reference frame using normalised mutual information [?], and applying each transformation consistently to the paired velocity volume,

$$\mathbf{V}_t^{\text{corr}} = \mathbf{T}_t^{\text{rigid}} \circ \mathbf{V}_t^{\text{raw}}, \qquad t = 1, \ldots, T, \tag{1}$$

followed by a vector-aware reorientation of the velocity components. To bring the 7 T volumes into the geometric space of the 3 T acquisitions, a temporal maximum-intensity projection $\text{MIP}(\mathbf{x}) = \max_t M_t(\mathbf{x})$ was first computed for each magnitude series to obtain a static, high-contrast angiographic representation. A rigid alignment followed by an affine refinement, both optimised with normalised mutual information, defined the composite transformation $\mathbf{T}_{7T\to 3T}^{\text{affine}}$, which was then applied to the full 4D velocity volume so that

$$\mathbf{V}_{7T\to 3T}^{\text{reg}}(\mathbf{x}, t) = \mathbf{T}_{7T\to 3T}^{\text{affine}} \circ \mathbf{V}_{7T}^{\text{corr}}(\mathbf{x}, t), \tag{2}$$

establishing voxel-level correspondence between the paired 3 T and 7 T data as required for supervised learning.

Figure 2: Registration pipeline. Temporal MIPs of the paired 3 T and 7 T magnitude before and after the rigid + affine inter-scan registration; the checkerboard overlay illustrates the resulting voxel-level alignment.

Angiographic volume synthesis and vascular masking

A voxel-level vascular probability map was constructed by combining two complementary contrast sources. A 4D speed image was obtained from the VENC-scaled velocity vector,

$$S(\mathbf{x},t) = \sqrt{\left(\frac{u(\mathbf{x},t)}{\mathrm{VENC}}\right)^2 + \left(\frac{v(\mathbf{x},t)}{\mathrm{VENC}}\right)^2 + \left(\frac{w(\mathbf{x},t)}{\mathrm{VENC}}\right)^2}, \tag{3}$$

and projected over the cardiac cycle as $\bar{S}(\mathbf{x}) = \max_t S(\mathbf{x},t)$. The temporal MIP of the magnitude and $\bar{S}$ were each min–max normalised and their voxel-wise product $\mathcal{A}(\mathbf{x}) = \widehat{\mathrm{MIP}}(\mathbf{x}) \cdot \hat{S}(\mathbf{x})$ defined a synthetic angiographic volume in which voxels that are both bright in magnitude and fast in flow are reinforced while static tissue and background noise are suppressed.

The Circle-of-Willis (CoW) mask used throughout this work was generated on the 7 T magnitude data, taking advantage of their substantially higher signal-to-noise ratio. For every subject, an initial foreground estimate was obtained by adaptive intensity thresholding of the 7 T maximum-intensity projection with connected-component cleanup, then refined by an ensemble nnU-Net inference and finally corrected by an expert reviewer on a slice-by-slice basis. The resulting manually corrected mask defined the gold-standard CoW segmentation in the 7 T geometry; transferring it to the 3 T frame via the registration above yielded the binary mask $\mathcal{M}(\mathbf{x})$ used as the supervision and evaluation region.

Two nnU-Net models with the residual-encoder *ResEnc-M* backbone [**?**] were then trained on these corrected labels with the skeleton-recall loss formulation introduced by the TopCoW-2024 winning solution [**?**, **?**]. The objective combined Dice, skeleton-recall, and cross-entropy terms,

$$\mathcal{L}_{\mathrm{seg}} = w_D\,\mathcal{L}_{\mathrm{Dice}} + w_S\,\mathcal{L}_{\mathrm{SkelRec}} + w_C\,\mathcal{L}_{\mathrm{CE}}, \qquad w_D = w_S = w_C = 1, \tag{4}$$

where the skeleton-recall term is computed on the fly from a differentiable topological skeletonisation of the ground-truth mask. Both 2D and 3D-full-resolution configurations were trained over five cross-validation folds and combined as a 2D + 3D ensemble at inference. The first model operates on the temporal maximum projection of the native 3 T magnitude (mean cross-validated Dice 0.83), and a second model was trained on the same projection after isotropic 0.5× downsampling (mean Dice 0.76), matching the input geometry of the super-resolution task introduced below. The same ensemble is used twice in this work: first, at training time, to propagate the 7 T-derived gold-standard mask through the 3 T, 0.5× 3 T, and 7 T-in-3 T volumes that form the supervised pairs; and second, at evaluation time, to re-segment each predicted 4D Flow volume so that its CoW geometry can be compared against the subject-specific 7 T reference mask.

Figure 3: Angiographic volume synthesis and vascular masking. The synthesised angiographic field $\mathcal{A}(\mathbf{x})$ combines magnitude and speed evidence; the gold-standard CoW mask is derived from the 7 T magnitude via a hybrid threshold, bootstrap nnU-Net, and manual correction, and a ResEnc-M U-Net trained with the skeleton-recall loss propagates this mask to the 3 T and 0.5×3 T inputs and to the predicted volumes used in evaluation.

Network architecture: CBAM-DNS and CBAM-SupRes

The core predictive backbone is an adapted SR4DFlowNet [**?**, **?**], a 3D fully convolutional residual network inspired by enhanced residual architectures for super-resolution [**?**], augmented with a Convolutional Block Attention Module (CBAM) [?] inserted between the low-resolution residual stack and the upsampling stage. We instantiate the same backbone in two task-specific forms used throughout the results: **CBAM-DNS** for same-resolution 3 T→7 T domain translation ($r = 1$) and **CBAM-SupRes** for joint domain translation and twofold super-resolution ($r = 2$).

The model accepts six scalar input channels per voxel — the three VENC-normalised velocity components (u, v, w) and their magnitude-weighted counterparts (u_m, v_m, w_m) — and internally derives three auxiliary fields that summarise the local kinematics and phase-contrast signal:

$$S = \sqrt{u^2 + v^2 + w^2}, \qquad M = \sqrt{u_m^2 + v_m^2 + w_m^2}, \qquad \mathrm{PCMR} = M \cdot S. \tag{5}$$

Velocity and physical channels are processed in two dedicated convolutional streams of two 3×3×3 layers with 64 feature maps, symmetric padding, and ReLU activation: a phase stream operating on (u, v, w) to extract flow-kinematic features, and a phase-contrast stream operating on (PCMR, M, S) to extract signal-intensity features. Their feature maps are concatenated and fused by a 1×1×1 projection followed by a 3×3×3 refinement convolution.

The fused representation is propagated through $N_{LR} = 8$ pre-activation residual blocks at the input resolution [?],

$$\mathbf{F}^{(l+1)} = \mathbf{F}^{(l)} + f\left(\mathbf{F}^{(l)}; \Theta^{(l)}\right), \tag{6}$$

after which the CBAM module sequentially applies channel and spatial attention to re-weight feature responses before upsampling. Channel attention is obtained from global average- and max-pooled descriptors processed by a shared two-layer multilayer perceptron, while spatial attention is computed from the channel-wise pooled feature maps and a 7×7×7 convolution; the two attention maps are applied multiplicatively in sequence. Spatial upsampling is then performed by trilinear interpolation with scale factor r, followed by $N_{HR} = 4$ high-resolution residual blocks operating at the target resolution. The network terminates in three independent two-layer convolutional heads that emit the reconstructed velocity components $(\hat{u}, \hat{v}, \hat{w})$, together with an additional head that predicts the reconstructed magnitude $\hat{M}$. Within this common backbone, CBAM-DNS uses $r = 1$ and trains under the same-resolution domain-translation setting, whereas CBAM-SupRes uses $r = 2$ and absorbs both the cross-field-strength translation and the spatial upsampling into a single forward pass.

Figure 4: Network architecture underlying CBAM-DNS ($r = 1$) and CBAM-SupRes ($r = 2$). The phase and phase-contrast streams are fused, refined by $N_{LR} = 8$ residual blocks, re-weighted by a CBAM channel-and-spatial attention module, upsampled by trilinear interpolation with factor r, and refined again by $N_{HR} = 4$ high-resolution residual blocks before three velocity heads and an optional magnitude head emit the reconstructed volume.

Loss function and evaluation metrics

Training minimises a composite, mask-aware loss that jointly penalises intraluminal velocity error, magnitude error, and texture irregularity in the extraluminal background,

$$\mathcal{L} = \mathcal{L}_{\mathrm{vel}} + \lambda_M \, \mathcal{L}_{\mathrm{mag}} + \lambda_{\mathrm{TV}} \, \mathcal{L}_{\mathrm{TV}}. \tag{7}$$

For each predicted velocity component $\hat{q} \in \{\hat{u}, \hat{v}, \hat{w}\}$ and its VENC-normalised target $q^* = q/\mathrm{VENC}$, the per-voxel squared error $\epsilon_q(\mathbf{x}) = (\hat{q}(\mathbf{x}) - q^*(\mathbf{x}))^2$ is split into a masked fluid term and a complementary background term,

$$\mathcal{L}^q_{\mathrm{fluid}} = \frac{\sum_{\mathbf{x}} \epsilon_q(\mathbf{x}) \, \mathcal{M}(\mathbf{x})}{\sum_{\mathbf{x}} \mathcal{M}(\mathbf{x}) + \varepsilon}, \qquad \mathcal{L}^q_{\mathrm{bg}} = \frac{\sum_{\mathbf{x}} \epsilon_q(\mathbf{x}) \, (1 - \mathcal{M}(\mathbf{x}))}{\sum_{\mathbf{x}} (1 - \mathcal{M}(\mathbf{x})) + \varepsilon}, \tag{8}$$

and aggregated as $\mathcal{L}_{\mathrm{vel}} = \sum_{q\in\{u,v,w\}}(\mathcal{L}^q_{\mathrm{fluid}} + \lambda_{\mathrm{bg}}\,\mathcal{L}^q_{\mathrm{bg}})$ with background weight $\lambda_{\mathrm{bg}} = 0.1$, reflecting the clinical priority of accurate intraluminal velocity quantification. $\mathcal{L}_{\mathrm{mag}}$ is the analogous masked mean-squared error on the per-frame min–max-normalised magnitude with magnitude weight $\lambda_M = 1$. To suppress spurious high-frequency texture in extraluminal regions, an isotropic total-variation penalty [?] is applied exclusively to voxel pairs that lie outside the mask along each Cartesian axis,

$$\mathcal{L}_{\mathrm{TV}} = \sum_q \sum_{d\in\{x,y,z\}} \frac{\sum_{\mathbf{x}} |\hat{q}(\mathbf{x}+\mathbf{e}_d) - \hat{q}(\mathbf{x})|\,\Omega(\mathbf{x},d)}{\sum_{\mathbf{x}} \Omega(\mathbf{x},d) + \varepsilon}, \tag{9}$$

where $\Omega(\mathbf{x},d) = (1-\mathcal{M}(\mathbf{x}))\,(1-\mathcal{M}(\mathbf{x}+\mathbf{e}_d))$ is the joint outside indicator and $\lambda_{\mathrm{TV}} = 10^{-5}$ provides a gentle smoothness prior without over-regularising the extraluminal signal.

Image-reconstruction quality was assessed by VENC-normalised mean-squared error (MSE) of the predicted velocity field. The intraluminal velocity error was computed inside the CoW mask both as a total

$$\mathrm{MSE}_{\mathrm{vel}} = \frac{\sum_{\mathbf{x}} \sum_{q\in\{u,v,w\}} (\hat{q}(\mathbf{x}) - q^*(\mathbf{x}))^2\,\mathcal{M}(\mathbf{x})}{\sum_{\mathbf{x}} \mathcal{M}(\mathbf{x}) + \varepsilon} \tag{10}$$

and per component

$$\mathrm{MSE}_q = \frac{\sum_{\mathbf{x}} (\hat{q}(\mathbf{x}) - q^*(\mathbf{x}))^2\,\mathcal{M}(\mathbf{x})}{\sum_{\mathbf{x}} \mathcal{M}(\mathbf{x}) + \varepsilon}, \qquad q \in \{u,v,w\}, \tag{11}$$

and an analogous extraluminal quantity — the background MSE $\mathrm{MSE}_{\mathrm{bg}}$ — was obtained by substituting $\mathcal{M}$ with $1-\mathcal{M}$ in the numerator and denominator. Both metrics are reported in units of 10^{-3} in the VENC-normalised range.

Vascular-geometry fidelity was evaluated by re-segmenting each predicted 4D Flow volume with the same ResEnc-M ensemble described above and comparing the resulting binary CoW mask P to the subject-specific 7 T reference mask R in the 3 T geometry. Volumetric overlap was quantified by the Dice score,

$$\mathrm{Dice}(P,R) = \frac{2|P\cap R|}{|P| + |R|}. \tag{12}$$

Boundary agreement was quantified in physical units from triangulated surfaces S_P and S_R extracted by marching cubes. Letting $d(p,S_R) = \min_{r\in S_R}\|p-r\|_2$ denote the directed point-to-surface distance, the symmetric mean surface distance is

$$\mathrm{sMSD}(P,R) = \tfrac{1}{2}\left(\frac{1}{|S_P|}\sum_{p\in S_P} d(p,S_R) + \frac{1}{|S_R|}\sum_{r\in S_R} d(r,S_P)\right), \tag{13}$$

and the worst-case boundary mismatch is captured by the Hausdorff distance,

$$\mathrm{HD}(P,R) = \max\left(\max_{p\in S_P} d(p,S_R),\ \max_{r\in S_R} d(r,S_P)\right). \tag{14}$$

Dice is reported as higher-is-better in $[0,1]$; sMSD and HD are reported in millimetres and lower-is-better. Together, these metrics quantify both the volumetric overlap and the boundary fidelity of the reconstructed vascular geometry that the velocity field implies.

Figure 5: Loss partition and evaluation regions. The CoW mask $\mathcal{M}$ separates fluid voxels (where $\mathcal{L}_{\text{vel}}$ and $\mathcal{L}_{\text{mag}}$ are evaluated and intraluminal MSE is reported) from extraluminal voxels (where $\mathcal{L}_{\text{TV}}$ and background MSE are computed). Geometry metrics are obtained from the predicted and reference surfaces S_P and S_R.

Data preparation, training, and inference

Prior to network input, all velocity components were normalised by the global VENC,

$$\tilde{q} = q/\text{VENC}_{\text{global}}, \qquad \text{VENC}_{\text{global}} = \max(\text{VENC}_u, \text{VENC}_v, \text{VENC}_w), \tag{15}$$

yielding values approximately in $[-1, 1]$, and magnitude images were rescaled to $[0, 1]$ by per-frame min–max scaling [?]. In both task configurations the reconstruction target was the registered 7 T velocity and magnitude volume after masking by $\mathcal{M}$, $\mathbf{V}^*(\mathbf{x}, t) = \mathbf{V}_{7T}^{\text{reg}}(\mathbf{x}, t)\,\mathcal{M}(\mathbf{x})$, confining supervision to the haemodynamically relevant intraluminal region. CBAM-DNS was trained at native 3 T resolution ($r = 1$), with $\mathbf{V}_{3T}^{\text{reg}}$ as input and $\mathbf{V}^*$ as target, so that the network simultaneously bridges the SNR gap between the two field strengths, corrects residual contrast differences arising from distinct hardware and reconstruction chains, and recovers finer vascular detail visible only at 7 T. CBAM-SupRes was trained with a spatially downsampled input $\mathbf{V}_{3T}^{\text{LR}}(\mathbf{x}, t) = \mathcal{Z}_{0.5}[\mathbf{V}_{3T}^{\text{reg}}(\mathbf{x}, t)]$, obtained by isotropic trilinear downsampling, and the same masked 7 T target, combining $r = 2$ upsampling and domain translation in a single end-to-end pass.

Each task was trained in two complementary input strategies that are reported side by side in the results. In the *synthetic-noise* strategy, the network sees the original 3 T (or downsampled 3 T) volume with synthetic non-vascular noise injected into the extraluminal region during training, and operates without a mask at inference time. In the *masked-input* strategy, the predicted CoW segmentation is used to zero the extravascular signal of the 3 T input prior to the network for both training and inference; no noise augmentation is then applied. The synthetic-noise strategy targets mask-free deployment and emphasises robustness to background patterns, whereas the masked-input strategy emphasises geometric fidelity by removing the extraluminal domain shift altogether.

Two augmentation strategies were applied online. Vector-aware geometric augmentation drew random 90° rotations from the planes $\{xy, xz, yz\}$ with multiplicities $\{1, 2, 3\}$ at probability 0.5, and the three velocity components were permuted and sign-corrected as a proper vector field while the mask and magnitude were rotated as scalar fields. Synthetic non-vascular noise augmentation (used in the synthetic-noise strategy only) injected per-batch amplitudes drawn from one of several statistical distributions (zero-mean normal, Student-t, Laplace, skew-normal, or generalised normal) into the extraluminal region of the input,

$$\tilde{\mathbf{V}}^{\text{LR}}(\mathbf{x}) = \mathbf{V}^{\text{LR}}(\mathbf{x}) + \eta(\mathbf{x})\,(1 - \mathcal{M}(\mathbf{x})), \tag{16}$$

encouraging mask-focused learning and preventing the network from overfitting to specific noise patterns in the background.

Training was conducted on three-dimensional patches of size 16^3 voxels in the input space, which corresponded to 32^3 voxels in the output space for the $\times 2$ configuration [?]. For each case, 64 patches were drawn per epoch with one cardiac frame selected uniformly at random per sample; 16 deterministic centre patches were used for validation, and an optional minimum-mask-coverage criterion ensured that each sampled patch contained a sufficient proportion of vascular voxels. Optimisation used Adam [?] with an initial learning rate of 2×10^{-4} and weight decay 5×10^{-7}. A ReduceLROnPlateau scheduler monitored the validation loss with a patience of ten epochs and halved the learning rate (floor 10^{-6}), and an early-stopping criterion with patience twenty epochs, complemented by an overfitting detector that terminated training when the training loss kept decreasing while the validation loss increased for fifteen consecutive epochs, prevented over-fitting. Models were trained with mini-batches of four samples; the checkpoint attaining the lowest validation loss was retained for inference.

All results are reported under *leave-one-out cross-validation* (LOOCV) at the patient level. With $n = 7$ subjects, seven independent folds were trained: in each fold, one subject was held out as the test case, one as the validation case used for model selection and early stopping, and the remaining five formed the training set; no subject appeared in more than one of these subsets. The metrics reported in the results are mean $\pm$ standard deviation across the seven held-out subjects.

Figure 6: Task configurations and patient-level leave-one-out cross-validation. (a) $\times 1$ domain translation and $\times 2$ joint domain translation and super-resolution share the same backbone and target. (b) Seven LOOCV folds; one subject is held out as test and one as validation in every fold.

At inference time, full 4D volumes were reconstructed with an overlap-aware sliding-window strategy [**?**, **?**]. For each time frame the volume was partitioned into overlapping 16^3 patches with 25% overlap, and predicted patches were recombined by Gaussian-weighted blending at overlapping boundaries,

$$\hat{\mathbf{V}}(\mathbf{x}) = \frac{\sum_k w_k(\mathbf{x})\, \hat{\mathbf{V}}_k(\mathbf{x})}{\sum_k w_k(\mathbf{x})}, \tag{17}$$

where $w_k(\mathbf{x})$ is a Gaussian window centred on patch k, eliminating patch-edge discontinuities. The reconstructed volumes were masked by $\mathcal{M}$ to constrain the output to the CoW region for the intraluminal metrics, and the predicted velocity components were finally rescaled to physical units via $\hat{q}_{\text{phys}} = \hat{q} \cdot \text{VENC}$. For the geometry evaluation, the reconstructed magnitude (and, when unavailable, the synthesised angiographic volume of Figure 3) was passed through the ResEnc-M segmentation ensemble to produce the predicted binary CoW mask P that was subsequently compared against the subject-specific 7 T reference R to obtain Dice, sMSD, and HD.

Figure 7: Inference and qualitative reconstruction on a representative held-out subject. Velocity vector maps, voxel-wise error maps, and 3D streamlines for the $\times 1$ and $\times 2$ configurations, together with the re-segmented predicted CoW surface overlaid on the 7 T reference surface used to compute Dice, sMSD, and HD.